

Exercise Sheet Week 10.5

Exercise 1. Find

$$\lim_{x \rightarrow 0} \frac{\sin(x^3)}{\sin x + \cos x - \log(1+x) - \exp(x^3)}$$

Exercise 2. (i) State Taylor's remainder theorem.

(ii) Apply Taylor's remainder theorem to show that for all $x > 0$ there is a c , where $0 < c < x$ such that $e^{-x^2} = 1 - x^2 + \frac{x^4}{4!}g(c)$ where $g(c) = (16c^4 - 48c^2 + 12)e^{-c^2}$.

Exercise 3. Use the Maclaurin series for $\sinh x$ and Taylor's Remainder Theorem to approximate $\sinh(0.5)$ to a four decimal place accuracy.

Exercise 4. Determine the Maclaurin expansion of $\arctan(x)$ and hence a series approximation of $\arctan(1) = \frac{\pi}{4}$. Estimate the maximum error of this approximation using the expansion up to and including order 5 terms.